Breaking and Defending Convolutional Neural Networks

An Empirical Study of Adversarial Attacks and Defenses on MNIST

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Executive Summary

Robustness evaluation of deep learning models against adversarial attacks.

- **Motivation:** Deep learning models are highly vulnerable to imperceptible adversarial perturbations.
- **Main Idea:** Evaluate white-box attacks (FGSM, PGD) against pre-processing (Blur) and structural defenses (Adversarial Training).
- **Key Result:** Iterative attacks (PGD) easily bypass standard models. True robustness requires PGD Adversarial Training, reducing attack success from 100% to 16.5%.

The Threat Landscape

Analyzing Adversarial Vulnerabilities in Neural Architectures

- **Adversarial Examples:** Mathematically calculated noise forcing confident misclassifications.

FGSM

Fast Gradient Sign Method: A fast, single-step gradient attack using the linear part of the loss function to generate perturbations.

PGD

Projected Gradient Descent: Powerful, iterative FGSM constrained within an l_infinity norm ball. Considered the universal first-order adversary.

Defense Mechanisms

Mitigating adversarial perturbations through pre-processing and native training

Easy Defense: Gaussian Blurring

Pre-processing denoising step designed to neutralize small-magnitude perturbations.

Acts as a low-pass filter to disrupt high-frequency adversarial noise.

Effective against weak, pixel-level attacks like simple FGSM.

Minimal computational overhead during inference.

Realistic Defense: Adversarial Training

Incorporate adversarial examples directly into the loss function during the training phase.

Trains the network natively on a continuous stream of adversarial examples.

Fundamentally shifts and smooths decision boundaries.

Provides robust generalization against stronger, iterative attacks (e.g., PGD).

Methodology & Approach

Experimental setup and robustness evaluation metrics

Dataset
`MNIST (Grayscale, 28x28)`

Model Architecture
`Custom CNN (2 Conv, 2 Linear, Dropout)`

Evaluation Metric
`ASR - % of correct predictions flipped`

Perturbation Bound
`epsilon = 0.3 (Aggressive)`

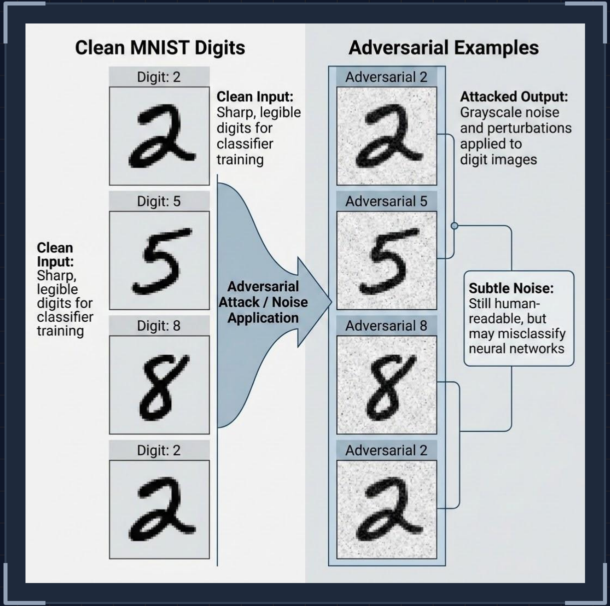
> *"The experimental framework prioritizes high-intensity adversarial pressure to test structural limits."*

Visual Evidence of Adversarial Perturbation

The disparity between human perception and neural network classification.

Clean Input

High-confidence classification of standard MNIST digits (2, 5, 8).



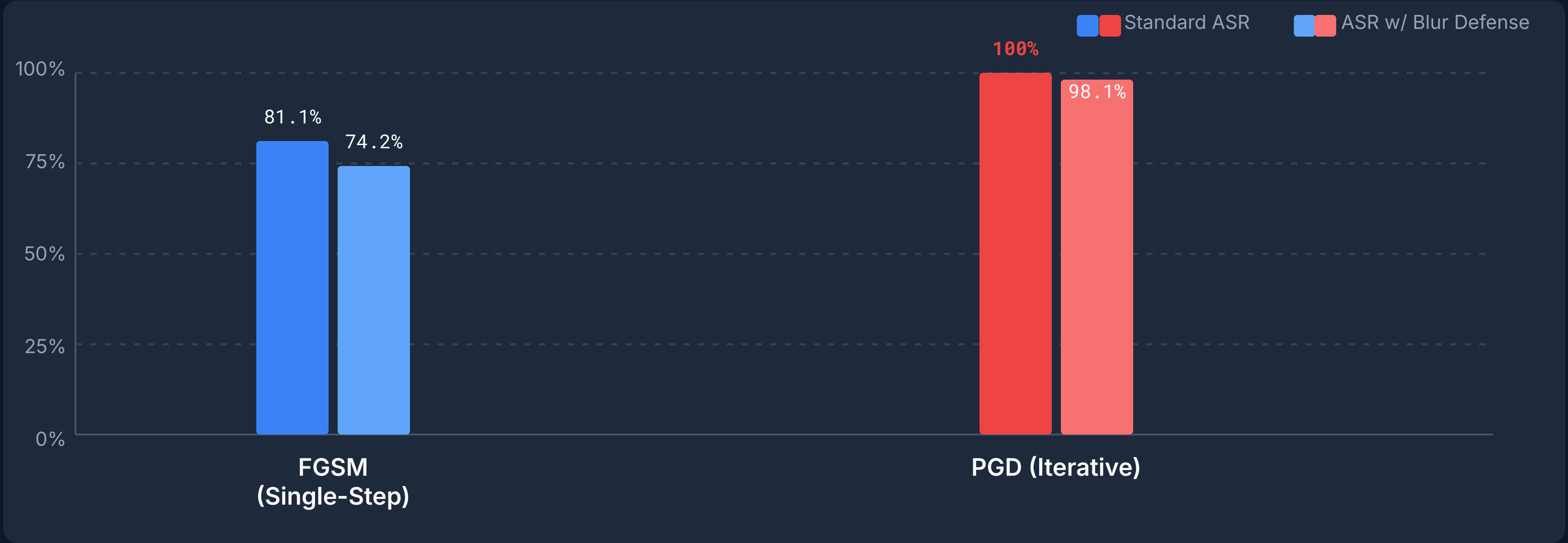
Adversarial Output

Subtle noise perturbations leading to targeted misclassification.

> *"Visual difference is minimal/noisy to the human eye. Mathematical difference completely shatters model confidence."*

Results: Baseline & Easy Defense

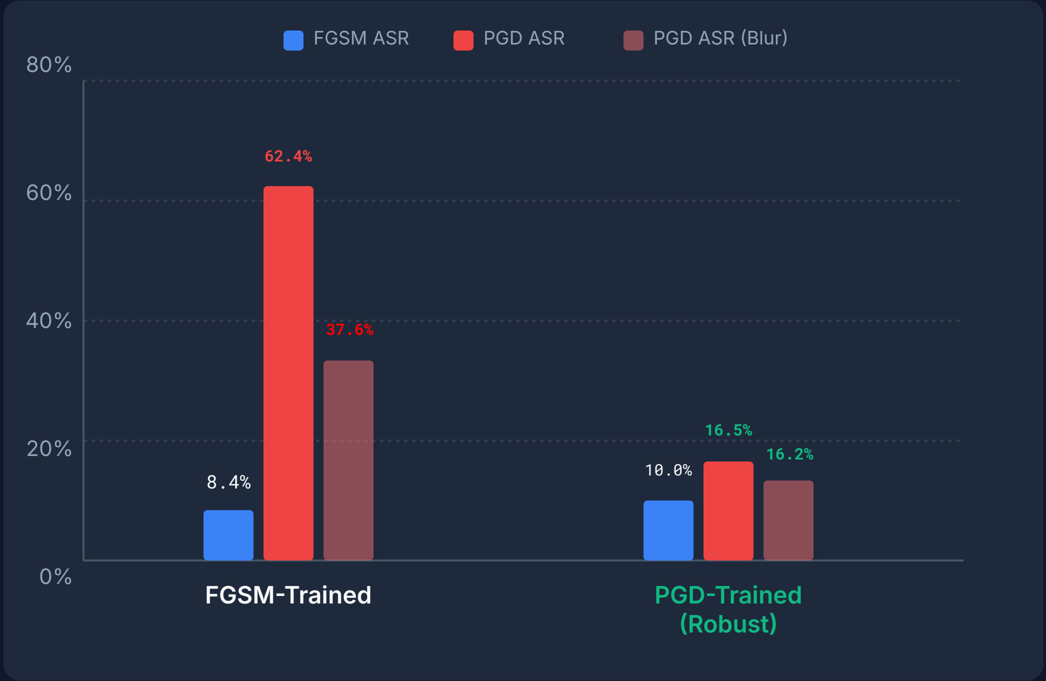
Standard Model Clean Accuracy: **98.69%**



Observation: Blur defense fails against optimal iterative attacks.

Results: Defense Evaluation

Comparative Robustness Metrics on MNIST Dataset



Clean Accuracy Performance

FGSM: 97.73% | **PGD:** 95.60%

False Security: FGSM-trained models collapse under iterative PGD attacks. **Ultimate Defense:** PGD-AT crushes PGD down to 16.50% ASR, rendering Blur obsolete.

Conclusion & The Robustness Trade-off

Balancing security and performance in adversarial environments

- **Iterative > Single-Step:** PGD easily finds optimal paths to cross decision boundaries where simpler methods fail.
- **Pre-processing Limits:** Gaussian Blurring cannot save defenseless models; it only masks the noise without fixing structural vulnerabilities.
- **The Trade-off:** True structural robustness requires PGD Adversarial Training. It drops standard accuracy slightly (~98% to ~95%) but provides massive security gains.

Adversarial training is the only proven method for achieving empirical robustness. The resulting accuracy penalty is a necessary cost for model reliability in high-stakes security applications.